

Food Microbiome Exposure

Ecosystem Stability and Personalized Diet-Microbiome
Associations

Team members:

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PI Planning: 4-Week Roadmap

Sprint 1

Data profiling, repository architecture, and baseline category mapping.

Sprint 2

CURRENT

To evaluate the impact of Food Microbiome Exposure (FME) on gut microbiome stability and diversity, daily food logs were mapped to the CFMD database to calculate individual FME scores for statistical analysis. Furthermore, we began an initial assessment of different models to evaluate which would best fit our work.

Sprint 3

Finalizing a working machine learning model, including hyperparameter tuning and producing a comprehensive summary report of the project's findings.

Roles Assignment

	Sprint1	Sprint2	Sprint3
Master Scrum / PO	Linoy	Maya	Ayelet
Lead Tech	Maya	Linoy	Ritaj
Scientists Data	Ritaj,Ayelet	Ritaj,Ayelet	Maya,Linoy

FME Framework & Data Integration

Objective

Investigate whether exposure to food-associated microbes influences gut microbiome diversity and stability.

Johnson et al. 2019

34 participants

566 fecal samples

Daily dietary logs

Gut microbiome sequencing

Demographic and nutrition data

cFMD

3,393 food samples

5,057 microbial taxa

Food-associated microbiome profiles

Final Analytical Set

475 matched fecal samples

samples with BOTH diet +
microbiome data

Insert a simple pipeline diagram:

Human Diet + Gut Microbiome Data

+

cFMD Food Microbiome Database

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Food Category Mapping

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Microbial Richness Weights

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FME Score Calculation

cFMD Weight Construction

Goal

Quantify microbial richness across food categories.

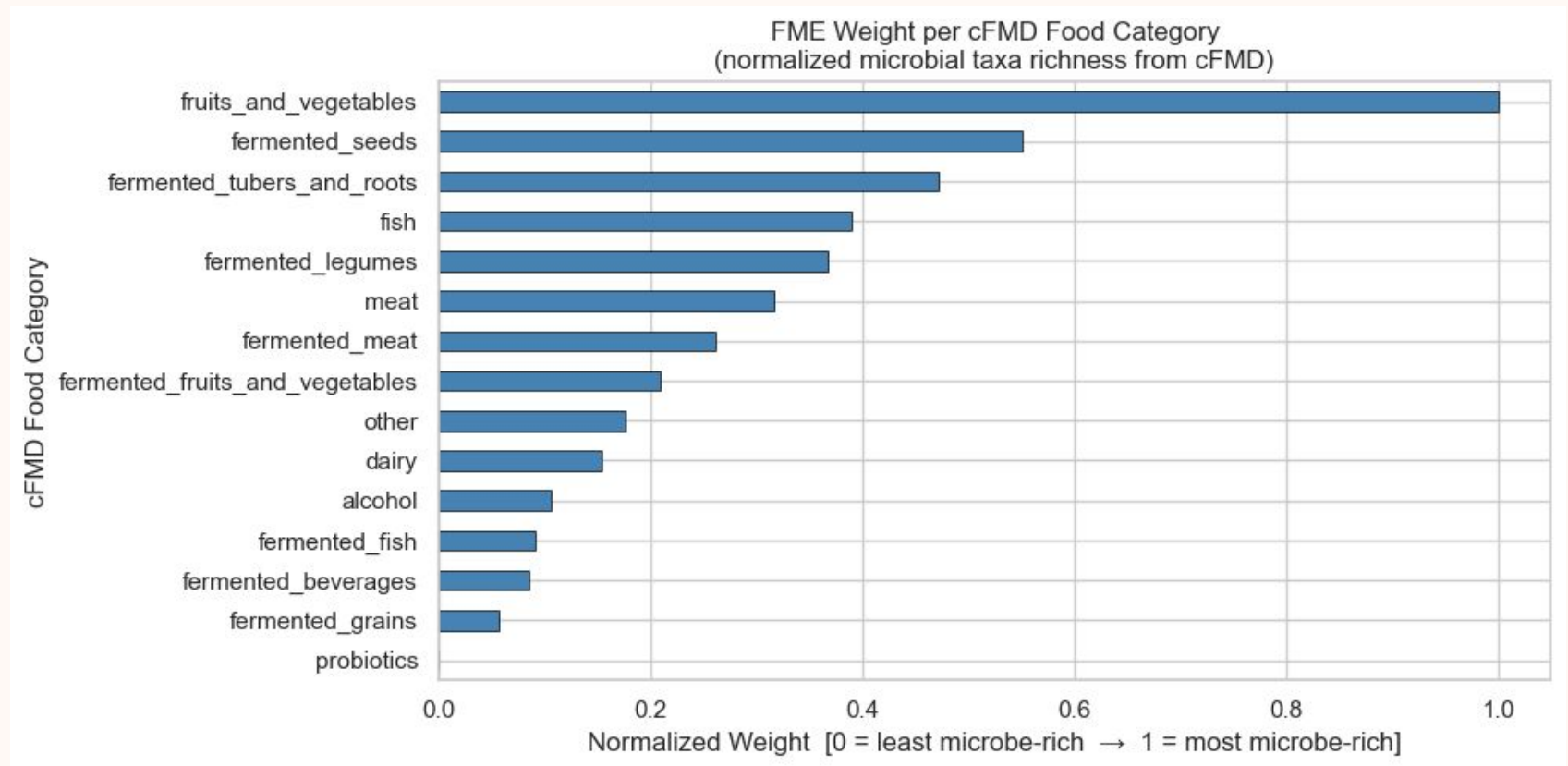
Method

Counted detected microbial taxa per food sample
Calculated mean richness per food category
Normalized weights to range 0–1

Main Findings

Fruits & vegetables showed the highest richness
Fermented foods showed elevated microbial diversity
Richness was used instead of abundance

FME Weight per cFMD Food Category



Foods with higher microbial richness contributed more strongly to the final FME score.

Diet-to-cFMD Mapping & FME Calculation

Daily FME Formula

$$FME_{sample} = \sum_{food\ items} consumption_{food,sample} \times w_{cfmd_category(food)}$$

Goal

Translate dietary intake into quantitative microbial exposure scores.

Mapping Strategy

Extracted L1 food categories from dietary taxonomy

Mapped dietary categories to corresponding cFMD food categories

Assigned microbial richness weights to each mapped category

Unmapped food items received weight = 0

Bottom Pipeline Visualization

Raw Diet Taxonomy

↓

L1 Category Extraction

↓

cFMD Category Mapping

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Microbial Weight Assignment

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Daily FME Score Calculation

Key Findings

- 475 daily FME scores computed
- Mean FME = 73.3
- Range = 0–331

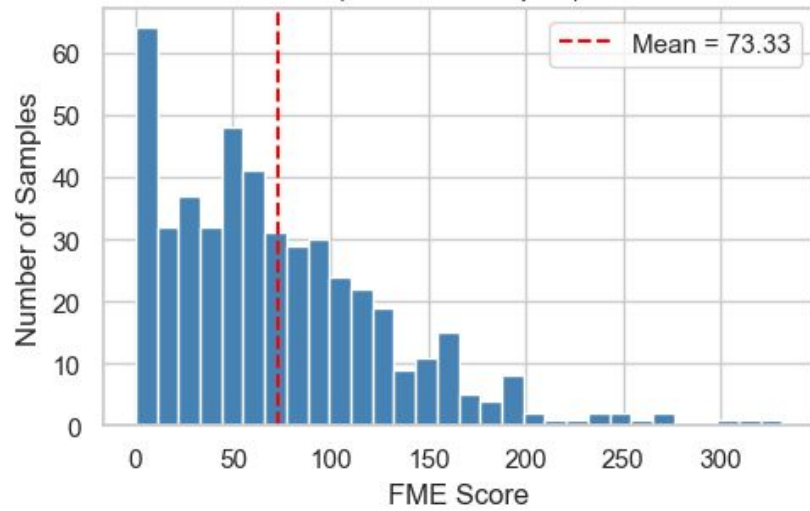
interpretation

Higher FME scores indicate greater exposure to diverse food-associated microbes.

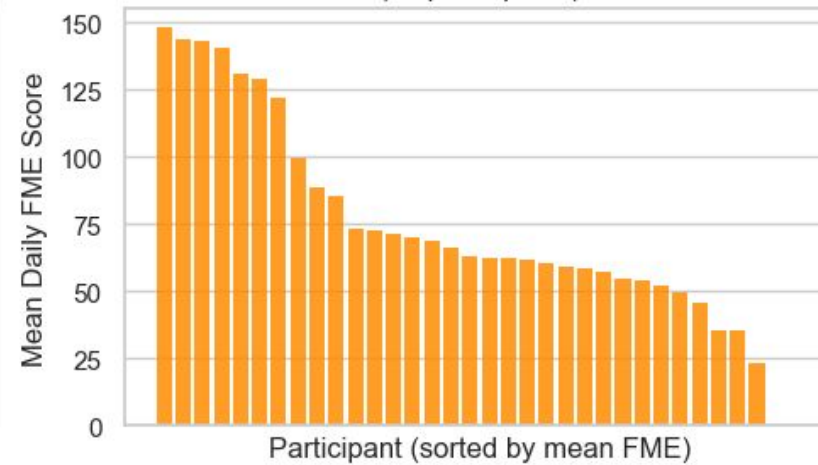
Diet-to-cFMD Mapping & FME Calculation

Food Microbiome Exposure (FME) Score — Overview

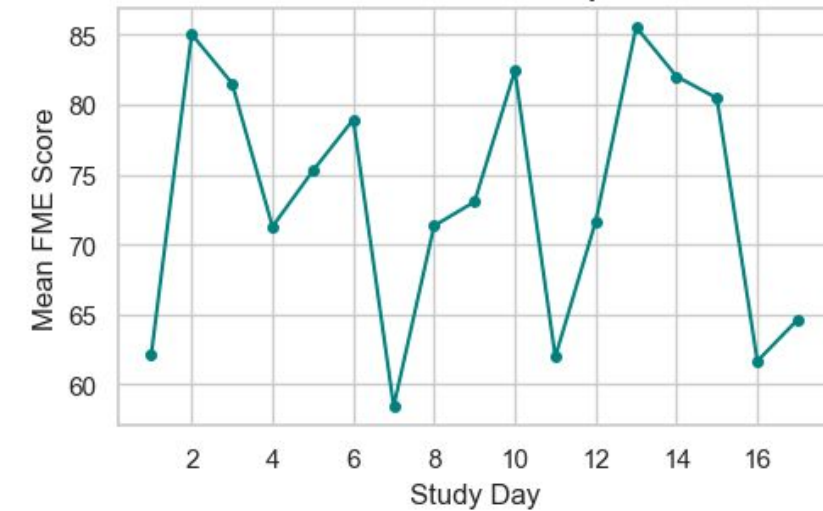
Daily FME Score Distribution
(475 fecal samples)



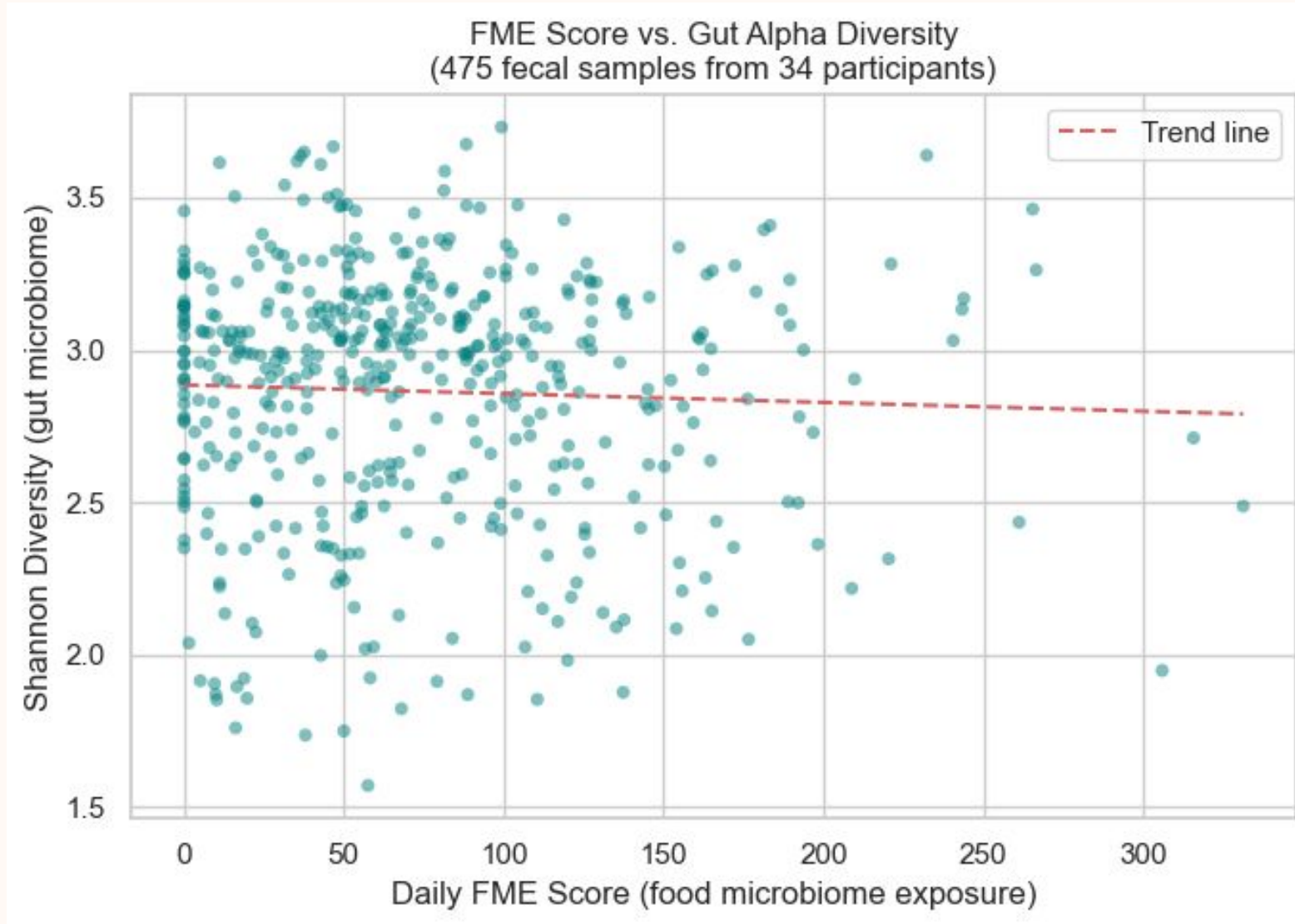
Mean FME Score per Participant
(34 participants)



Mean FME Score Over Study Timeline



FME vs Gut Microbiome Diversity



Diet-to-cFMD Mapping & FME Calculation

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Bottom Pipeline Visualization

Raw Diet Taxonomy



L1 Category Extraction



cFMD Category Mapping



Microbial Weight Assignment



Daily FME Score Calculation

Where

consumption = dietary intake amount

weight = microbial richness score derived from cFMD

interpretation

Higher FME scores indicate greater exposure to diverse food-associated microbes.

Baseline ML Model: Binary Classification

The goal:

Predict whether each sample has Low or High microbiome diversity based on dietary and participant level features.

Target definition:

Shannon diversity was split into two classes using the median:

- Low diversity:
Shannon $\leq$ median
- High diversity:
Shannon $>$ median

Features used:

FME score, KCAL, FIBE, CARB, TFAT, Age, BMI

Models tested:

Logistic Regression, Random Forest, XGBoost

Validation:

GroupKFold cross validation was used to avoid splitting samples from the same participant between train and test sets.

Results & Observations

Model performance:

The binary classification models showed weak predictive performance, with ROC-AUC close to random classification. The best relative model was Random Forest, with Accuracy = 0.521 and ROC-AUC = 0.600. XGBoost achieved the highest Accuracy = 0.560, but its ROC-AUC was slightly lower at 0.574.

Best model:

No model showed a strong advantage, suggesting that the binary target may not capture the full relationship between diet and microbiome diversity.

Next step:

Reformulate the task as regression and predict the continuous Shannon diversity value directly.

Initial feature insights:

Feature importance analysis was used to identify which dietary and participant level variables contributed most to the prediction.

Main limitation:

Splitting Shannon diversity into Low/High classes loses information, because Shannon diversity is naturally a continuous biological measure.

Statistical Analysis & Feature Selection

- Analyzed relationships between variables and microbiome characteristics.
- Performed **Correlation Analysis** to identify significant relationships between variables.
- Selected relevant features for the models using statistical analysis methods.
- Identified variables with the greatest impact on the results.
- Presented graphs and tables summarizing the main findings.

Key Outcome:

Successfully reduced the number of variables and focused on the most informative features, improving model performance and interpretability.



Model Evaluation & Main Findings

Results & Observations



- השוואת ביצועי המודלים שנבחנו
- הערכת ביצועים באמצעות Accuracy ו-ROC-AUC
- ניתוח Feature Importance לזיהוי המשתנים המשפיעים ביותר
- הפקת תובנות לגבי הקשר בין תזונה להרכב המיקרוביום
- תיעוד המגבלות והאתגרים של הדאטה

:Main Conclusion

המודלים הצליחו לזהות דפוסים משמעותיים בנתונים, כאשר מספר קטן של משתנים תרם לרוב יכולת החיזוי.

Model Evaluation & Main Findings

Results & Observations

- Compared the performance of the selected models.
- Evaluated model performance using **Accuracy** and **ROC-AUC** metrics.
- Conducted **Feature Importance** analysis to identify the most influential variables.
- Derived insights regarding the relationship between nutrition and microbiome composition.
- Documented the limitations and challenges of the dataset.

Main Conclusion:

The models successfully identified meaningful patterns in the data, demonstrating that a relatively small number of variables can provide strong predictive value.



Association Testing Results

- Applied OLS regression to evaluate associations between nutritional variables and Shannon diversity.
- No nutritional variable showed a statistically significant association ($p > 0.05$).
- The model explained a small proportion of the variance ($R^2 = 0.027$).

Main Finding:

No strong association was found between the examined dietary variables and gut microbiome diversity.

OLS Regression Results						
Dep. Variable:	shannon_diversity	R-squared:	0.027			
Model:	OLS	Adj. R-squared:	0.014			
Method:	Least Squares	F-statistic:	2.152			
Date:	Tue, 09 Jun 2026	Prob (F-statistic):	0.0465			
Time:	20:31:00	Log-Likelihood:	-247.17			
No. Observations:	475	AIC:	508.3			
Df Residuals:	468	BIC:	537.5			
Df Model:	6					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	2.7853	0.061	45.544	0.000	2.665	2.906
fme_score_daily	-6.297e-05	0.000	-0.156	0.876	-0.001	0.001
KCAL	0.0002	0.000	1.326	0.185	-9.51e-05	0.000
PROT	1.476e-05	0.001	0.015	0.988	-0.002	0.002
TFAT	-0.0023	0.001	-1.582	0.114	-0.005	0.001
CARB	-0.0001	0.001	-0.191	0.849	-0.002	0.001
FIBE	-0.0043	0.002	-1.715	0.087	-0.009	0.001
Omnibus:	29.179	Durbin-Watson:	0.518			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	33.191			
Skew:	-0.645	Prob(JB):	6.20e-08			

Questions?

Thank you for your
attention.

Git:

[MayaGold42/Gut-Microbiome-Project_year3](#)